

Echoes of Antiquity: An Interactive Installation for Guqin Culture Heritage Using Mid-Air interaction and Generative AI

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Abstract

With the development of mid-air interaction, the digital preservation and interactive learning of intangible cultural heritage (ICH) have become increasingly significant. However, the intrinsic significance of the cultural symbols embedded within numerous ICH remains largely obscure to the general public. This paper introduces "Echoes of Antiquity", an interactive installation that utilizes Leap Motion for gesture recognition and generative AI for image processing to vividly illustrate the symbolic elements of Guqin culture, thus bridge the existing chasm in public understanding and appreciation of Guqin's rich cultural heritage. Specially, our system utilizes Leap Motion to capture gestures and deliver AI-generated images as feedback, thereby enhancing the understanding and retention of the Guqin's cultural heritage through the seamless integration of motion and visual cues.

CCS Concepts

• Human-centered computing → interaction design.

Keywords

Guqin Art, Mid-air interaction, Generative AI, Cultural Preservation, Interactive Design

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1 INTRODUCTION

The Guqin, one of China's oldest plucked string instruments, is integral to Chinese culture, reflecting its profound historical and philosophical traditions. However, the rapid development of modern society has posed numerous challenges to the inheritance of Guqin art, including the loss of performance skills and a decline in interest among the younger generation. Despite scholars' efforts to digitize Guqin culture [Zhang et al. 2015], there remains a deficiency in the integration of visual and auditory elements for

cultural symbols. One of the key texts on Guqin culture is *Tai Gu Yi Yin*, a Song Dynasty monograph that uses illustrations of animal dynamics to mimic gesture silhouettes, aiding readers in memorization. To capture the essence of form resemblance, our approach combines gesture-based interactivity with AI-driven image generation, immersing participants in the depth of Chinese cultural heritage and facilitating an embodied cognition that engages them kinesthetically, as well as visually and auditorily.

2 RELATED WORK

The digital preservation of ICH has garnered considerable academic interest [Isa et al. 2018], particularly due to the Guqin's historical and cultural significance in China, making it a key subject of study. For instance, Yu et al. developed interactive systems like VirGuqin and MRGuqin to improve public accessibility to the Guqin and facilitate learning [Yu et al. 2021]. Additionally, Lyu et al. used 3D hand pose estimation to capture the intricate fingering techniques vital for Guqin performance [Lyu et al. 2021]. However, despite these efforts to enhance accessibility and engagement, a research gap remains concerning the exploration of the cultural symbols and narratives central to the Guqin's heritage. Recently, Generative AI has demonstrated potential in the preservation of ICH by enabling innovative expressions of cultural elements through artificial intelligence [Xie 2022]. Nonetheless, the literature on the application of gesture-based interactivity combined with Generative AI for the preservation and dissemination of ICH, particularly concerning ancient instruments like the Guqin, remains limited.

3 DESIGN AND IMPLEMENTATION

In the "Echoes of Antiquity" project, participants are guided to explore Guqin finger techniques and their philosophy. By following instructions to mimic Guqin motions, our mid-air interaction system recognizes their movements. In response, we use Generative AI to stylistically render illustrations from the ancient text *Tai Gu Yi Yin*, activating the hand shapes that represent the visual symbol of Guqin.

3.1 Design concept

To catalyze the spread of Guqin culture, this work is crafted to entice those unfamiliar with the instrument to delve into its profound implications. Our innovation combines Guqin music with visual symbols from ancient texts, creating an immersive audio-visual experience. By using generative AI, we blend these symbols with Chinese ink painting to offer a unique representation, allowing the Guqin's cultural narrative to resonate in a contemporary context.

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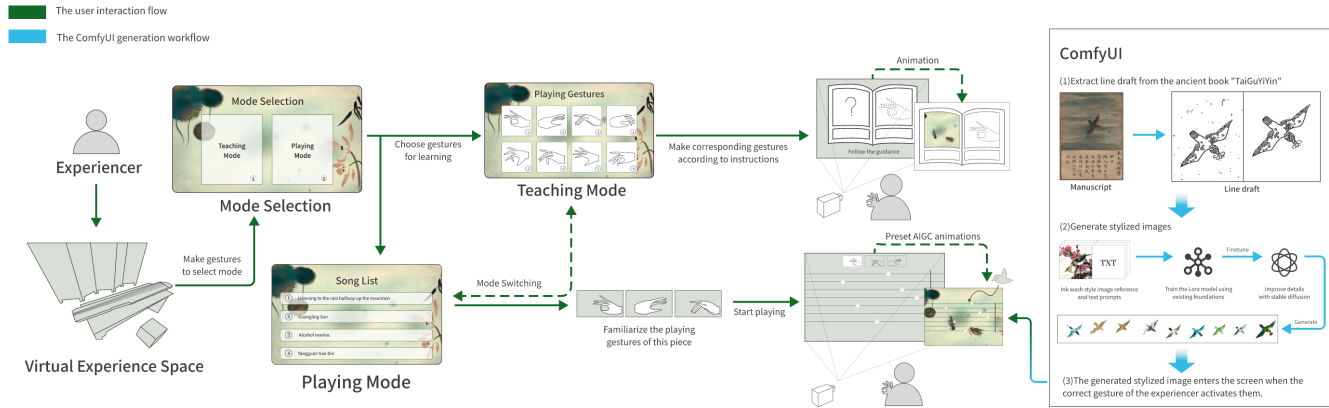


Figure 1: System Illustration. The system consists of two flows: the user interaction flow and the ComfyUI generation workflow

3.2 Mid-air interaction and feedback

Utilizing Leap Motion sensors integrated with Unity and Ultra-leap, our system creates an interactive installation that revitalizes the Guqin's cultural heritage. The system's core lies in its gesture recognition technology, which captures users' hand movements and compares them against an array of traditional Guqin playing gestures. Our design comprises two primary modes: a teaching mode that facilitates learning through intuitive symbols and guided practice, and a playing mode that allows exploration and performance of curated songs. The Leap Motion sensor enables gesture-controlled interactions, fostering an accessible and engaging interface without physical contact.

3.3 Visual symbol creation using generative AI

Notably, we harness an innovative Generative AI workflow to revitalize traditional Guqin culture through contemporary digital expressions. Our approach begins with the extraction of animal line drawings from the classical text *Tai Gu Yi Yin*, which serve as foundational elements for our digital renditions. These outline drawings are processed through a pre-trained LoRA (Low-Rank Adaptation) model, fine-tuned on a dataset of 500 Chinese ink paintings to authentically capture the style and texture of this artistic tradition. Key parameters include a learning rate of 0.001, a batch size of 16, and training conducted over 50 epochs. To further enhance visual fidelity and stylistic consistency, we incorporate Stable Diffusion to transform line drawings into illustrations that capture the distinctive features of Chinese ink painting while preserving their original silhouettes. Additionally, to ensure swift responsiveness, our system integrates a cache of 50 pre-processed AI images to guarantee real-time visual responses to user interactions, enhancing the overall experience.

4 CONCLUSION

As we delve into the realm of Guqin cultural heritage, the integration of Generative AI in our interactive installation represents a key innovation in our educational strategy. We introduce "Echoes of Antiquity", an interactive experience crafted to engage the general

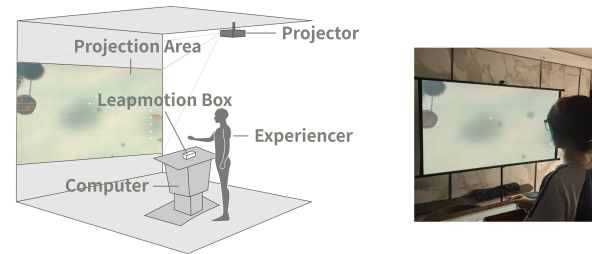


Figure 2: Installation layout and site situation.

public and foster an appreciation for Guqin culture, making it accessible to those with little prior exposure or interest in the art form. This approach democratizes learning by removing barriers to physical instruments and providing immediate, interactive feedback that is culturally resonant. Future efforts will focus on expanding the installation's reach and enhancing its impact on cultural education.

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